Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

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Introduction to Data Science

Assignment # 03

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34. Title Page Along with GitHub Repository Link

GitHub Repository Link:

https://github.com/Qaim-Ali727/AQI-Data-Science-Project/tree/main

35. Introduction

Air quality pollution has become a serious environmental and public health concern across the world. Harmful pollutants such as PM2.5, PM10, nitrogen dioxide, sulfur dioxide, and carbon monoxide negatively affect both humans and ecosystems. The Air Quality Index (AQI) is commonly used to measure pollution severity and indicate whether air conditions are safe or harmful.

This project uses the **Global Urban Air Quality Index Dataset (2015–2025)** to analyze pollution trends, study pollutant behavior, classify AQI categories using machine learning algorithms, and identify hidden structures through clustering and dimensionality reduction techniques.

36. Dataset Description

The dataset contains air quality and weather-related information collected from multiple international cities over a ten-year period.

Feature	Description
Total Records	3,660 entries
Total Columns	13 variables
Time Period	2015 to 2025
Data Condition	No missing values or duplicate records were identified
Coverage	Data collected from 9 different countries
Target Variables	AQI and AQI Category

37. Data Cleaning Steps

Proper preprocessing was performed before applying visualization and machine learning techniques.

Step 1 – Duplicate Removal

The dataset was checked for repeated entries, and duplicate records were removed if found.

Step 2 – Missing Value Handling

Numerical missing values were replaced using the mean of each respective column to maintain consistency.

Step 3 – Date Formatting

The Date column was converted into date time format to support time-based analysis.

Step 4 – Feature Engineering

Year and Month columns were extracted from the Date feature to analyze yearly and seasonal AQI patterns.

Step 5 – Data Type Validation

Pollution and weather-related columns were converted into numeric data types where necessary.

Step 6 – AQI Category Creation

AQI numerical values were transformed into categorical classes ranging from **"Good"** to **"Hazardous"** using standard AQI ranges.

38. Basic Statistics

The following statistics summarize the major air quality measurements across all 3,660 records.

Statistic	AQI	PM2.5	PM10	Humidity (%)	Wind (m/s)	Temp (°C)
Mean	164.64	126.38	~125.0	50.66	7.79	~22.5
Minimum	30	5.10	~5.0	10	0.50	~-10
Maximum	300	250.00	~250	90	15.00	~45

39. Exploratory Data Analysis with Charts

Several visualizations were created using Matplotlib and Seaborn to better understand pollution patterns and feature relationships.

AQI Category Distribution

Most observations belong to the **Moderate** and **Unhealthy** AQI classes, while **Good** air quality records are comparatively fewer.

Country-Based Pollution Analysis

Countries such as **India, China, and Egypt** showed relatively higher AQI values compared to countries like **Australia and the United States**.

AQI Trend Over Time

The yearly trend analysis suggests that pollution levels generally increased between **2015 and 2025**.

PM2.5 Relationship with AQI

A very strong positive correlation (**r ≈ 0.95+**) was observed between PM2.5 and AQI, indicating that PM2.5 is one of the strongest contributors to pollution levels.

Weather Impact

Wind speed showed a negative relationship with AQI because stronger winds help reduce pollutant concentration in the atmosphere.

40. KNN Results

The **K-Nearest Neighbors (KNN)** algorithm was trained using scaled numerical features with an 80/20 train-test split.

Parameter	Result
Best Accuracy Achieved	Approximately 25%
Best k Value	7
Observation	KNN showed weak performance due to overlapping AQI categories

The model produced lower accuracy because several AQI categories overlap with each other, making classification difficult for distance-based algorithms like KNN.

41. Naive Bayes Results

Gaussian Naive Bayes was applied using the same features and target variable.

Parameter	Result
Accuracy Achieved	38.66%
Correct Predictions	283 out of 732 test samples
Performance	Better than KNN

Naive Bayes performed better than KNN because its probabilistic approach handled feature distributions more effectively.

42. K-Means Clustering Results

K-Means clustering was applied after standardizing the numerical features and removing categorical labels.

Cluster	Average AQI	Interpretation
Cluster 0	~90	Cleaner air conditions with stronger wind movement
Cluster 1	~165	Moderate pollution levels under average conditions
Cluster 2	~245	Severe pollution with high PM2.5 concentration

The clustering process successfully identified meaningful pollution categories and highlighted the connection between stagnant air and high pollution.

43. PCA Visualization

Principal Component Analysis (PCA) was used to reduce multiple numerical features into two principal components.

Feature	Result
Variance Explained	21.4%
Components Used	2 Principal Components
Purpose	Data visualization and dimensionality reduction

Although PCA captured only a portion of the total variance, the scatter plot still showed visible separation between cleaner and highly polluted groups.

44. Final Comparison Table

Method	Task	Parameter	Result	Verdict

KNN	Classification	k = 7	25% Accuracy	Weakest
Naive Bayes	Classification	Gaussian NB	38.66% Accuracy	Best Classifier
K-Means	Clustering	k = 3	3 Clusters	Effective
PCA	Dimensionality Reduction	2 Components	21.4% Variance	Useful for Visualization

45. Conclusion

The analysis of **3,660 air quality records** demonstrates the growing environmental challenge caused by air pollution worldwide. PM2.5 was identified as the most influential pollutant affecting AQI values.

Among the supervised learning models, **Naive Bayes** achieved better performance than KNN for AQI classification. Clustering techniques successfully separated the dataset into low, medium, and highly polluted groups. PCA also helped visualize complex environmental patterns in two dimensions.

Future improvements may include advanced machine learning models such as **Random Forest, XGBoost, or time-series forecasting methods** for improved prediction accuracy.

46. GitHub Repository Link

https://github.com/Qaim-Ali727/AQI-Data-Science-Project/tree/main

47. References

1. **Scikit-learn Documentation (2024)**
2. **World Health Organization (WHO) Air Quality Guidelines (2021)**
3. **Kaggle – Global Urban Air Quality Index Dataset (2015–2025)**

58. What is AQI?

AQI stands for **Air Quality Index**. It is a measurement system used to describe how clean or polluted the air is and whether it can affect human health.

59. Why is AQI Important?

AQI is important because poor air quality can cause respiratory diseases, environmental damage, and serious health problems for sensitive populations.

60. Which City or Country Has the Highest AQI in the Dataset?

The maximum AQI value of **300** was observed in **São Paulo, Brazil**, while the minimum AQI value of **30** was recorded in **Los Angeles, USA**. Countries such as **Brazil, India, and Egypt** generally showed higher pollution levels compared to **Australia and the USA**.

61. Which Pollutant Seems Most Related to AQI?

PM2.5 showed the strongest relationship with AQI because increases in PM2.5 concentrations significantly increased pollution levels.

62. What Cleaning Steps Were Required?

The dataset preprocessing included:

- Duplicate checking
- Missing value handling
- Date conversion
- Feature engineering
- Validation of numeric columns
- AQI category creation

63. What Patterns Did You Observe from the Charts?

- Most cities fall within Moderate and Unhealthy AQI categories
- PM2.5 has a strong positive relationship with AQI
- Stronger wind speeds are associated with cleaner air
- Pollution levels generally increased from 2015 to 2025

64. Which Algorithm Performed Better: KNN or Naive Bayes?

Naive Bayes performed better with an accuracy of **38.66%**, compared to KNN which achieved approximately **25% accuracy**. The probabilistic nature of Naive Bayes handled overlapping AQI classes more effectively.

65. What Did the K-Means Clusters Show?

The clustering process identified three pollution groups representing:

- Cleaner air conditions
- Moderate pollution
- Heavily polluted environments

66. What Did PCA Help You Understand?

PCA reduced high-dimensional pollution data into two principal components, making it easier to visualize relationships and identify separation between cleaner and polluted regions.

67. What Are the Limitations of Your Analysis?

The analysis was limited by:

- Relatively low classification accuracy
- Simplified machine learning models
- Limited parameter tuning
- Lack of advanced temporal forecasting techniques